

Estimated Calculation Spread sheet

Integrated Energy calculation for EV Charging Stations

Power Generation							PCC - Point of common coupling			ESS - Charging	
Power Source	Voltage [V]	Current [I]	% of Generation/day	Required Power	Power [P]- KW/hr	Power/day	PCC will Logically controlled on aavailability of source			No.of batteries	40
Solar	430	16.7	60	72	43	1032	Voltage - [V]	430	Output Power/day in KW	Voltage	12
ESS	480	16.7	30	80	24	576	Current - [I]	17		Amper hour - capacity	100
Grid	435	20.0	10	87	9	209	Power - [W] - KW/h	75	1800 Gen /day	Power / charge - Watt / hr	48000
Total Power Generation per day in KW/24 hr						1817	Power Availability	1.17	1772 use / day	Power / one cycle charge - KW	48

Fast Charging Station Load								
Cars	Voltage (V)	Power (kW)	Battery Capacity (kWh)	SOC Available	Required Current	Charging Time in h	Unit charge	Cost
1 - Tata Nexon - EV	400	30.2	30.2	20	75.5	2.47	10	745.83
2- MG ZS EV	400	50	44.5	30	125	1.49	10	743.75
3-Tata Trigo EV	400	16.8	26	50	42	4.40	10	739.58
4- Mahindra eVerito	400	70	60	10	175	1.07	10	747.92
Total Power		167			Total Time	9	Total Revenue	2977
Total Power		1670	KW power can charge / day		Total Time	/ 96 hr	Total Revenue	26794

Battery Swapping Station				
Total Batteries	30			
Under maintance	0			
Available batteries	30			
Voltage	12			
Current	10			
Time / Battery in hr	5			
No of Cycle per day	3			
Power/Battery - KW	0.12			
Total Station KW	18			
Total Energy Required for Full Swapping Station per Day - KW	54			

Total Batteries charging /	90
Cost of charge per batter	30
Total Amout Per day	2700

Total Energy claculation				
Power sources	KW / day	KW / Month	KW / year	KW / 5 years
Required full Power From Grid	1,817	54,510	663,205	3,316,025
Solar	1,032	30,960	376,680	1,883,400
ESS	576	17,280	210,240	1,051,200
Grid	209	6,270	76,285	381,425
Total Power saving / Day	1,608	48,240	586,920	2,934,600
Total Generation Cost / Day 12rs / unit	21,804	654,120	7,958,460	39,792,300
Total Generation Cost Saving / Day 12rs / unit	19,296	578,880	7,043,040	35,215,200
Spending Amount	2,508	75,240	915,420	4,577,100

Revenue				
Source	Per day	Per Month	Per Year	Per 5 Years
Fast charging	26,794	803,813	9,779,719	48,898,594
Battery Swapping	2,700	81,000	985,500	4,927,500
Total Revenue	29,494	884,813	10,765,219	53,826,094
Spending Amount	2,508	75,240	915,420	4,577,100
Total Profit	26,986	809,573	9,849,799	49,248,994

